

Sarthak Bichhawa

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Summary

Senior Data Scientist (9+ years) building ML and analytics for marketplace, retail, and growth—assortment optimization, demand transfer, revenue attribution, and customer retention—including ~7% category revenue lift and ~\$45M/month in recovered partner revenue.

Experience

Microsoft, Redmond, WA | *Sr. Data & Applied Scientist*

Nov 2020 – Current

Revenue Growth & Attribution:

- Recovered and sustained ~\$45M/month in influenced partner revenue (<1% variance vs Finance, ~4× partner coverage) by rebuilding a 20+ component attribution pipeline with **multi-partner credit** replacing last-writer-wins—lifting partner-attributed influenced revenue >10% month-over-month.
- Collapsed monthly business-review reporting from a multi-day manual process to a governed ~5-minute pipeline (>95% effort reduction), now in regular leadership use.

Customer Retention & Health:

- Helped reverse churn/downtrend risk for **10+ critical customers**—each representing multi-millions in monthly Azure Storage revenue—by accelerating early detection and action on customer-impacting issues.
- Drove the **Mission-Critical Customers (MCC) health program** and a unified customer-health view across **8+ engineering/support teams**, cutting executive/customer readiness insight from **3–5 days to near real-time**—with ~50% fewer >10-day aging issues and ~16% faster time-to-mitigate.

Applied ML & Optimization:

- Enabled fleet-scale similarity search, segmentation, and embedding-drift anomaly detection across **millions of accounts** by designing an **account-behavior embedding model** (autoencoder + FAISS) that compresses each Azure Storage account into a vector fingerprint.
- Helped prevent **10+ Sev1/Sev2 incidents** with adaptive pre-release stress-testing for Azure Storage—thousands of controlled simulations per build prioritized by **multi-armed bandits** (60% fewer false positives).

AI-Assisted Insights:

- Architected an enterprise **LLM agent** that turns natural-language questions into governed live Kusto/ADX queries and PM-ready narratives, cutting customer-health time-to-insight from **days to <2 minutes** across 13 metric domains.

Walmart Labs, Bentonville, AR | *Senior Data Scientist*

Feb 2019 – Nov 2020

- Delivered ~7% category revenue lift across 50 stores with an end-to-end **planogram optimization** system, cutting planning cycles from weeks to hours.
- Built **product association + substitution analytics** (behavior tree + demand-transfer “universes”) to support assortment and replenishment decisions.
- Led **4 DS/analysts + 2 interns** to productionize a repeatable item-level placement workflow; earned a Spot Award.

Integral Ad Science, New York, NY | *Data Scientist Intern*

June 2018 – Aug 2018

- Built a PySpark contextual text-classification pipeline processing ~1M URLs/day across **846 categories** (88% accuracy); improved label quality via correlation analysis.

Mu Sigma Business Solutions, Bengaluru, India | *Data Scientist*

Oct 2015 – July 2017

- Estimated **customer lifetime value (CLTV)** with probabilistic/regression models; improved competitive reporting by unifying product IDs via fuzzy matching.

Education

University of Illinois at Chicago (UIC), Chicago, IL | *MS-Business Analytics*

Aug 2017 – Dec 2018

SGSITS, Indore, India | *B.E. in Computer Engineering*

Aug 2011 – July 2015

Skills

- Analytical Techniques:** Causal inference & experimentation, optimization & simulation, demand transfer & substitution analytics, segmentation & CLTV, online learning (multi-armed bandits), anomaly detection, generative & agentic AI
- Analytical Languages & Tools:** Python, SQL, Azure Data Explorer (Kusto), PySpark, R, FastAPI, Power BI, Tableau, Azure DevOps, Git
- Functional Areas:** Marketplace & pricing analytics, retail & merchandising (planogram & assortment optimization), supply/demand optimization, partner/revenue attribution, customer retention & growth, production ML